64. A bottle contains 20 litres of liquid A. 4 litres of liquid A is taken out of the it and replaced by same quantity of liquid B. Again 4 litres of the mixture is taken out and replaced by same quantity of liquid B. What is the ratio of quantity of liquid A to that of liquid B in the final mixture?

(a) 4 : 1 (b) 5 : 1
(c) 16 : 9 (d) 17 : 8

Ans. (c) : Remaining quantity of liquid (A) = $P\left(1-\frac{R}{P}\right)^n$

where P = initial quantity
$R \rightarrow$ Replace quantity
$n \rightarrow$ number of processes

$$= 20\left(1-\frac{4}{20}\right)^2$$

$$= 20\left(1-\frac{1}{5}\right)^2$$

$$= 20\times\left(\frac{4}{5}\right)^2 = 20\times\frac{16}{25}$$

Remaining quantity of liquid (A) $= \frac{64}{5}$ Litre

The, the quantity of B in mixture $= 20-\frac{64}{5}$

$= \frac{36}{5}$ litre

Required Ratio $= \frac{64}{5}:\frac{36}{5}$

$= 16 : 9$

65. The average score of a batsman after his 50th innings was 46.4. After 60th innings, his average score increases by 2.6. what was his average score in the last ten innings?

(a) 122 (b) 91
(c) 62 (d) 49

Ans. (c) : ∵ The average score of batsman after 50 innings - 46.4
∴ The total score of batsman after 50 innings
= 46.4 × 50
= 2320.0

∵ After 60 innings average score = 46.4 + 2.6
= 49.0

The total score after 60 innings = 49 × 60
= 2940

Then the average score of 10 innings $= \frac{2940-2320}{10}$

$= \frac{620}{10} = 62$

or

The average score of 10 innings = $46.4 + \frac{60\times2.6}{10}$

=46.4 + 15.6 = 62.0

66. As a result of 25% hike in the price of rice per kg, a person is able to purchase 6 kg less rice for ₹ 1,200. What was the original price of rice per kg?

(a) ₹ 30 (b) ₹ 40
(c) ₹ 50 (d) ₹ 60

Ans. (b) : Original Price $= \frac{a}{n}\times\frac{x}{(100\pm x)}$

Where a = amount
n = Less/More things
x = Increase or decrease percentage

$= \frac{1200}{6}\times\frac{25}{125}$

$= 200\times\frac{1}{5} = 40$ ₹/kg

67. What is the greatest length x such that $3\frac{1}{2}$ m and $8\frac{3}{4}$ m are integral multiples of x?

(a) $1\frac{1}{2}$ m (b) $1\frac{1}{3}$ m
(c) $1\frac{1}{4}$ m (d) $1\frac{3}{4}$ m

Ans. (d) : On taking H.C.F. of $3\frac{1}{2}$m, $8\frac{3}{4}$m

$\because 3\frac{1}{2} = \frac{7}{2}$m and $8\frac{3}{4} = \frac{35}{4}$m

are fraction

$\because$ H.C.F. of fraction $= \frac{\text{H.C.F. of Numerator}}{\text{L.C.M. of Denominator}}$

$= \frac{\text{A.C.F. of }(7,35)}{\text{L.C.M. of }(2,4)}$

$= \frac{7}{4}$

$= 1\frac{3}{4}$m

68. Consider the following data :

Year	Birthrate	Death rate
1911-1921	**48.1**	**35.5**
1921-1931	**46.4**	**36.3**
1931-1941	**45.2**	**31.2**
1941-1951	**39.9**	**27.4**
1951-1961	**41.7**	**22.8**
1961-1971	**41.1**	**18.9**
1971-1981	**37.1**	**14.8**

For which period was the natural growth rate maximum?

(a) 1911-1921 (b) 1941-1951
(c) 1961-1971 (d) 1971-1981

Ans. (d) :

∵ Natural growth rate = Birth rate – Death rate

The growth rate of 1911 – 1921 = 48.1 – 35.5
= 12.6

The growth rate of 1941 – 1951 = 39.9 – 27.4
= 12.5

The growth rate of 1961 – 1971 = 41.1 – 18 – 9
= 22.2

The growth rate of 1971 – 1981 = 37.1 – 14.8
= 22.3

Hence, The natural growth rate maximum was the period 1971 – 1981.

69. The recurring decimal representation 1.272727……is equivalent to

(a) 13/11
(b) 14/11
(c) 127/99
(d) 137/99

Ans. (b) : $1.272727..... = 1.\overline{27}$

$= 1+0.\overline{27}$

$= 1+\frac{27}{99}$

$= 1+\frac{3}{11}$

$= \frac{14}{11}$

70. What is the least four-digit number when divided by 3, 4, 5 and 6 leaves as remainder 2 in each case?

(a) 1012
(b) 1022
(c) 1122
(d) 1222

Ans. (b) : On taking L.C.M. 3, 4, 5 and 6.
The L.C.M. of (3, 4, 5, 6) = 2 × 2 × 3 × 5 = 60

∵ The least four digit number = 1000

then $\frac{1000}{60} \xrightarrow{\text{Remainder}} 40$

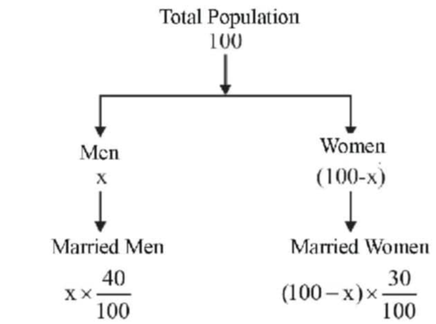
∴ The least four digit number which is completely divisible by 3, 4, 5 and 6 = 100 + (60 – 40)
= 1020

∵ Leaves remainder 2 in each case,
Hence, The required number = 1020 + 2
= 1022

71. In adult population of a city, 40% men and 30% women are married. What is the percentage of married adult population if no man marries more than one woman and no woman marries more than one man; and there are no widows and widowers?

(a) $33\frac{1}{7}\%$ (b) 34%
(c) $34\frac{2}{7}\%$ (d) 35%

Ans. (c) : Let total population = 100
and population of men = x
then population of women = (100 – x)



According to question

$$x\times\frac{40}{100} = (100-x)\times\frac{30}{100}$$

$$4x = 300-3x$$

$$7x = 300$$

$$x = \frac{300}{7}$$

then married men $= x\times\frac{40}{100} = \frac{300}{7}\times\frac{40}{100} \Rightarrow \frac{120}{7}$

married women $= (100-x)\times\frac{30}{100}$

$= \left(100-\frac{300}{7}\right)\times\frac{30}{100}$

$= \frac{400}{7}\times\frac{30}{100} \Rightarrow \frac{120}{7}$

Total married men and women $= \frac{120}{7}+\frac{120}{7} \Rightarrow \frac{240}{7}$

Married percentage $= \frac{\frac{240}{7}}{100} = \frac{240}{7}\%$

$= 34\frac{2}{7}\%$